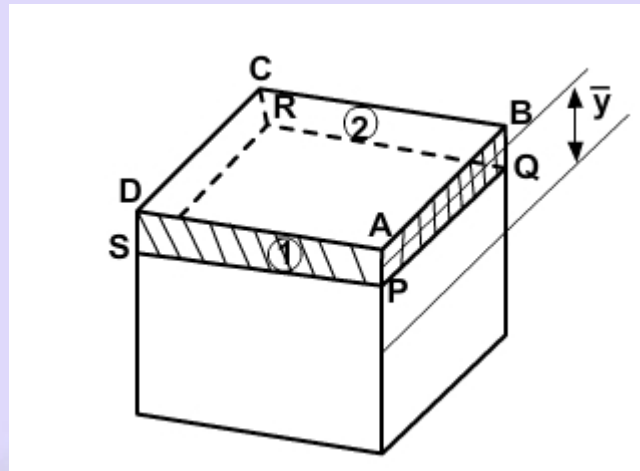


Shear Stresses in Beams

Shear force is related to change in bending moment between adjacent sections.



$$\frac{dM}{dx} = v$$

As shown in figure, consider a cut-out section from a beam.

At section 1 {Area PADS}

$$F_1 = \int_{A_1} \frac{-M_1 y}{I} dA = \frac{-M_1}{I} \int_{A_1} y dA = \frac{M_1 Q}{I}$$

where y is the distance of any fiber from the neutral axis.

$$Q = \int_{A_1} y \cdot dA \text{ - First moment of area about the neutral axis } F_2 = \frac{M_2 Q}{I}$$

if $M_2 = M_1 + dM$ and $F_2 = F_1 + dF$

$$\text{then } dF = \frac{dM}{I} Q$$

$$\text{Force per unit length, } \frac{dF}{dx} = \frac{dM}{dx} \frac{Q}{I}$$

This force per unit length is termed as the shear flow, q

$$\text{Substituting for } \frac{dM}{dx} = V ;$$

$$q = \frac{VQ}{I}$$

Here I is the moment of Inertia of the entire cross-sectional area around the neutral axis.

Shear Stress Formula for Beams

The shear stress formula is obtained by modifying the shear flow formula.

$$\frac{dF}{dx} = \frac{dM}{dx} \frac{Q}{I}$$

Shear Stress,

$$\begin{aligned}\tau &= \frac{1}{t} \frac{dF}{dx} \\ &= \frac{VQ}{It} \\ &= \frac{q}{t}\end{aligned}$$

(Refer figure)

